COMPARISON DIAGRAM OF TOPOI BETWEEN THE BIG FPPF SIMPLICIAL SITE AND THE SMALL ETALE SITE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the case-A assertions ('This is immediate from the construction', source lines 7266-7268) refer to the definitional form of the sites' coverings; the author's footnote on the etale-topology variant of the total site is preserved verbatim; this lemma underlies the descent theorems delivered as cards L46151 (ph version) and L46161 (etale version) and the ringed variant is card L46165.

Editorial note. The label lemma-compare-simplicial-objects-fppf-etale names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The hypercovering total-site construction over (Spaces/S)_fppf. Same chapter, Section \label{section-hypercovering}. Lines 7251-7256: (Spaces/U)_{fppf,total} as the total site of the simplicial object U in (Spaces/S)_fppf (footnote on the etale variant).
- The sites representing the small etale topoi. Same chapter, Section \label{section-simplicial-algebraic-spaces}. Lines 7256-7259: X_{spaces,etale} and U_{spaces,etale} may be used for the right-hand topoi.
- Both sites fall into case A of the simplicial-site situation. Same chapter, Situation \label{situation-simplicial-site} and \label{lemma-sr-when-fibre-products}. Lines 7261-7266: the case-A placement of (Spaces/U)_{fppf,total} and U_{spaces,etale}.
- The component morphisms of sites a_{U_n} and a_X. More on Cohomology of Spaces chapter, Section \label{spaces-more-cohomology-section-fppf-etale}. Lines 7268-7276: U_{n,spaces,etale} -> (Spaces/U_n)_fppf and X_{spaces,etale} -> (Spaces/X)_fppf are morphisms of sites, denoted a_{U_n} = epsilon_{U_n} pi_{u_n} and a_X = epsilon_X pi_X.
- Morphisms of simplicial sites compatible with augmentations. Same chapter, Remark \label{remark-morphism-augmentation-simplicial-sites} and \label{lemma-morphism-augmentation-simplicial-sites}. Lines 7277-7284: the componentwise data form a morphism of simplicial sites compatible

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with the augmentations, and the cited lemma produces the commutative diagram.

Source reference. Original reference 0D99: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1402–1402. Prerequisite pointer invoked at proof line(s) 7283; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WL: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5136–5136. Prerequisite pointer invoked at proof line(s) 7269; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D98: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1372–1372. Prerequisite pointer invoked at proof line(s) 7281; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 09A0: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 5012–5012. Prerequisite pointer invoked at proof line(s) 7246; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0DE7: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 6985–6985. Prerequisite pointer invoked at proof line(s) 7248; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 7266; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0DGE: sources/stacks-chow/tags.txt, source lines 16869–16869. Prerequisite pointer invoked at proof line(s) 7277; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let U be a simplicial algebraic space over S. Let $a : U \to X$ be an augmentation. There is a commutative diagram*

$$\begin{array}{ccc} Sh((Spaces/U)_{fppf,total}) & \xrightarrow[h]{} & Sh(U_{étale}) \\ {\scriptstyle a_{fppf}}\downarrow & & \downarrow{\scriptstyle a} \\ Sh((Spaces/X)_{fppf}) & \xrightarrow{h_{-1}} & Sh(X_{étale}) \end{array}$$

where the left vertical arrow is defined in Section 09A0 and the right vertical arrow is defined in Section 0DE7.

Proof. The notation $(\mathit{Spaces}/U)_{fppf,total}$ indicates that we are using the construction of Section 09A0 for the site $(\mathit{Spaces}/S)_{fppf}$ and the simplicial object U of this site[1]. We will use the sites $X_{spaces,étale}$ and $U_{spaces,étale}$ for the topoi on the right hand side; this is permissible see discussion in Section 0DE7.

Observe that both $(\mathit{Spaces}/U)_{fppf,total}$ and $U_{spaces,étale}$ fall into case A of Situation 09WE. This is immediate from the construction of $U_{étale}$ in Section 0DE7 and it follows from Lemma 09WL for $(\mathit{Spaces}/U)_{fppf,total}$. Next, consider the functors $U_{n,spaces,étale} \to (\mathit{Spaces}/U_n)_{fppf}$, $U \mapsto U/U_n$ and $X_{spaces,étale} \to (\mathit{Spaces}/X)_{fppf}$, $U \mapsto U/X$. We have seen that these define morphisms of sites in More on Cohomology of Spaces, Section 0DGE where these were denoted $a_{U_n} = \epsilon_{U_n} \circ \pi_{u_n}$ and $a_X = \epsilon_X \circ \pi_X$. Thus we obtain a morphism of simplicial sites compatible with augmentations as in Remark 0D98 and we may apply Lemma 0D99 to conclude. □

[1]We could also use the étale topology and this would be denoted $(\mathit{Spaces}/U)_{étale,total}$.